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Java Chat Application Using Socket Programming

Introduction

Communication systems are an essential component of modern software applications. Real-time messaging platforms allow users to exchange information instantly across networks. This project focuses on developing a simple chat application using Java that enables multiple clients to connect to a server and communicate in real time.

The system uses socket programming to establish connections between clients and the server. Each client communicates with the server using input and output streams, while the server manages multiple connections using multithreading. The chat interface is implemented using Java Swing to provide a simple graphical user interface for sending and receiving messages.

Abstract

The Java Chat Application is a networking-based system developed using Java Socket Programming and Swing GUI. The application enables multiple users to communicate with each other in real time through a client-server architecture. The server listens for incoming client connections and manages communication between clients using multithreading. The application supports group messaging, private messaging, and notifications when users join or leave the chat. This project demonstrates the implementation of networking concepts, concurrent programming, and graphical user interface development in Java.

Tools and Technologies Used

Technology	Purpose
Java	Core programming language
Socket Programming	Network communication
Multithreading	Handling multiple clients
Java Swing	Graphical user interface
Eclipse IDE	Development environment

Steps Involved in Building the Project

1. Created a server program that listens for client connections using ServerSocket.
2. Implemented a client program that connects to the server using Socket.
3. Developed a ClientHandler class to manage communication with each connected client.
4. Implemented multithreading to handle multiple users simultaneously.
5. Implemented message broadcasting to send messages to all connected clients.
6. Added private messaging functionality using the @username format.
7. Designed a graphical chat interface using Java Swing.
8. Tested the system by connecting multiple clients to the server and exchanging messages.

Conclusion

The Java Chat Application successfully demonstrates real-time communication using socket programming in Java. The system allows multiple users to connect to a server and exchange messages efficiently through a graphical interface. The project illustrates important networking concepts such as client-server architecture, multithreading, and real-time message broadcasting. This project helped enhance understanding of network communication, concurrent programming, and GUI development in Java.